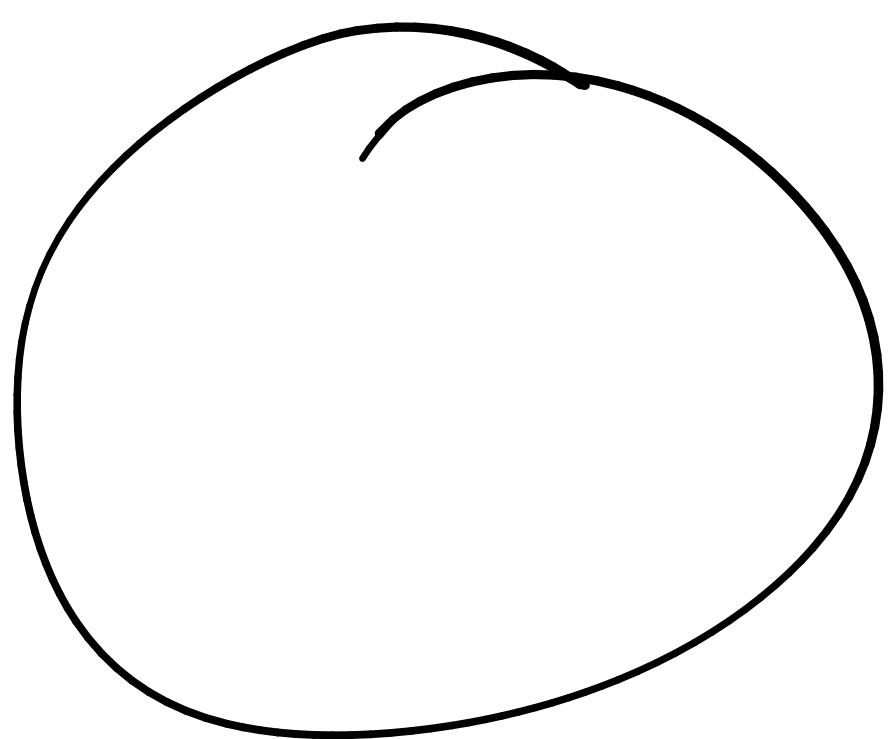


Cálculo

Libro de pre-cálculo.

Stewart

M
21:00-22:00



D
22:00-23:00

Página web

CG

+

VE

+ PDF'S

+ AI.

Algebra

3. POTENCIA CON EXPONENTE ENTERO

Resolver aplicando las propiedades de las potencias.

$(-)^{\text{par}} = (+)$
 $a^3 \cdot a^{-n} = 1, a \neq 0$
 $a^m \cdot b^n = (ab)^n$
 $(a^x)^y = a^{x \cdot y}$
 $\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}$

(1) $\left(\frac{2^{-4}}{2^{-2} - 2^{-3}} \right)^{-2}$
 (2) $\frac{(-3)^{-3} \cdot 15^{-1} \cdot (-25^{-2})^{-2} \cdot 5^{-3}}{[(-45)^{-2}]^2 \cdot 9^2 \cdot (-5)^4}$
 (3) $\frac{7 \cdot 2^{30}}{2^{32} + 2^{31} + 2^{30}}$
 (4) $\frac{(-3)^{-2} \cdot 2^{-3} \cdot \left(\frac{1}{72}\right)^{-1}}{(-3)^3 - \left[\left(-\frac{1}{2}\right)^{-2}\right]^2}$

(2) $\frac{(-3)^{-3} \cdot 3^{-2}}{3^8 \cdot 3^4} \cdot \frac{5^{-4} \cdot 5^8 \cdot 5^{-3}}{5^4 \cdot 5^4 \cdot 1}$
 $= -\frac{3^{-4}}{3^4} \cdot 5^4$
 $= -625$

$(5^{-2})^{-2}$

$2^{-3} = \frac{1}{2^3}$

$a^4 = \frac{1}{a^{-4}}$

$\frac{3^{-4}}{3^{12}} = 3^{-4-12} = 3^{-16}$

$a^{-n} = \frac{1}{a^n}$

$2^3 = \frac{1}{2^{-3}}$

¿Qué es un logaritmo?

Def

$$a = \log_b(b^a)$$

$\log_b(a)$

$$a = b$$
$$b > 0 \wedge b \neq 1$$

$$a^x = b \Leftrightarrow x = \log_a b$$

$$17) E = x^2 + x + 1 - 2 \log x + 3 \log(x+1)$$

$$(i) n \cdot \log_a b = \log_a(b^n)$$

$$E = x^2 + x + 1 + \log(x^{-2}) + \log(x+1)^3$$

$$(ii) \log_a x + \log_a y = \log_a(x \cdot y)$$

$$(iii) \log_a x - \log_a y = \log_a\left(\frac{x}{y}\right)$$

$$E = x^2 + x + 1 + \log(x^{-2}(x+1)^3)$$

$$E = \log(10^{x^2+x+1}) + \log(x^2(x+1)^3)$$

$$E = \log(10^{x^2+x+1} \cdot x^2(x+1)^3)$$

$$\log_a b^n = n \cdot \log_a b$$

$$\log_{a^n} b = \frac{1}{n} \cdot \log_a b$$

Dividir por un número equivale a multiplicar por su recíproco.

Cambio de base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

$c \in \mathbb{R}^+ \setminus \{1\}$

$$a = \log_b(b^a)$$

$$a = b^{\log_b(a)}$$

$$27 = 3 \cdot \left(\frac{3}{2}\right)^x \quad | \cdot \frac{1}{3}$$

$$9 = \left(\frac{3}{2}\right)^x \quad | \log_{(3/2)}()$$

$$\boxed{\log_{(3/2)} 9} = x$$

$\ln x$

$\log(x)$

$$\log_{(3/2)} 9 = \frac{\log 9}{\log(3/2)} \approx$$

$$= \frac{\ln 9}{\ln(3/2)} \approx$$

8 Sept

Fraccus

Poturus

Logarithmus

$$a^{\log_a b} = b$$

D29

$$(4) \frac{1}{1+\sqrt{2}-\sqrt{3}}$$

D3

$$\frac{N}{a\sqrt{b}} = \frac{\sqrt{b}}{a\sqrt{b}}$$

$$\text{Caso 1: } \frac{N}{\sqrt{a}} = \frac{N\sqrt{a}}{a}$$

$$\text{Caso 2: } \frac{N}{\sqrt[n]{a^k}} = \frac{N \cdot \sqrt[n]{a^{n-k}}}{a}$$

$$\text{Caso 3: } \frac{N}{\sqrt{a} \pm \sqrt{b}} \cdot \frac{(\sqrt{a} \mp \sqrt{b})}{(\sqrt{a} \mp \sqrt{b})} = \frac{N(\sqrt{a} \mp \sqrt{b})}{a-b}$$

$$- \frac{4}{5} \log(x+2) + 2 \log(x+1)$$

(2)

$$\Rightarrow \frac{1}{1+\sqrt{2}-\sqrt{3}} = \frac{(1+\sqrt{2})+\sqrt{3}}{(1+\sqrt{2})+\sqrt{3}} \cdot \frac{(1+\sqrt{2})-\sqrt{3}}{(1+\sqrt{2})-\sqrt{3}} = \frac{(1+\sqrt{2})+\sqrt{3}}{(1+\sqrt{2})^2-3} = \frac{(1+\sqrt{2})+\sqrt{3}}{1+2\sqrt{2}+2-3} = \frac{(1+\sqrt{2})+\sqrt{3}}{2\sqrt{2}}$$

$$2) \frac{2}{3} \log(x-2) - \frac{4}{5} \log(x+2) + 2 \log(x+1)$$

$$\log((x-2)^{\frac{2}{3}}) - \log((x+2)^{\frac{4}{5}}) + \log((x+1)^2)$$

$$\log((x-2)^{\frac{2}{3}} \cdot (x+2)^{-\frac{4}{5}} \cdot (x+1)^2) = \log\left(\frac{(x-2)^{\frac{2}{3}} \cdot (x+1)^2}{(x+2)^{\frac{4}{5}}}\right)$$

$$\log\left(\frac{(x-2)^{\frac{2}{3}} \cdot (x^2+2x+1)}{(x+2)^{\frac{4}{5}}}\right)$$

$$\log\left(\frac{(x-2)^{\frac{2}{3}} \cdot (x^2+2x+1)}{(x+2)^{\frac{4}{5}}}\right) \Rightarrow \sqrt[3]{(x-2)^2}$$